

Database Design Explanation Document

Normalization from Flat Table to Third Normal Form (3NF)

1. Overview

The original data source is a single flat (denormalized) table containing all oil and gas well-related information in one place. This design document describes the process and rationale for normalizing this flat table into a structured, relational database schema consisting of nine separate tables.

The normalization process follows the principles of Third Normal Form (3NF), eliminating data redundancy, ensuring data integrity, and improving maintainability. Where no natural unique identifier existed in the source data, a surrogate primary key was introduced and highlighted in the Entity-Relationship Diagram (ERD).

2. Normalization Rationale

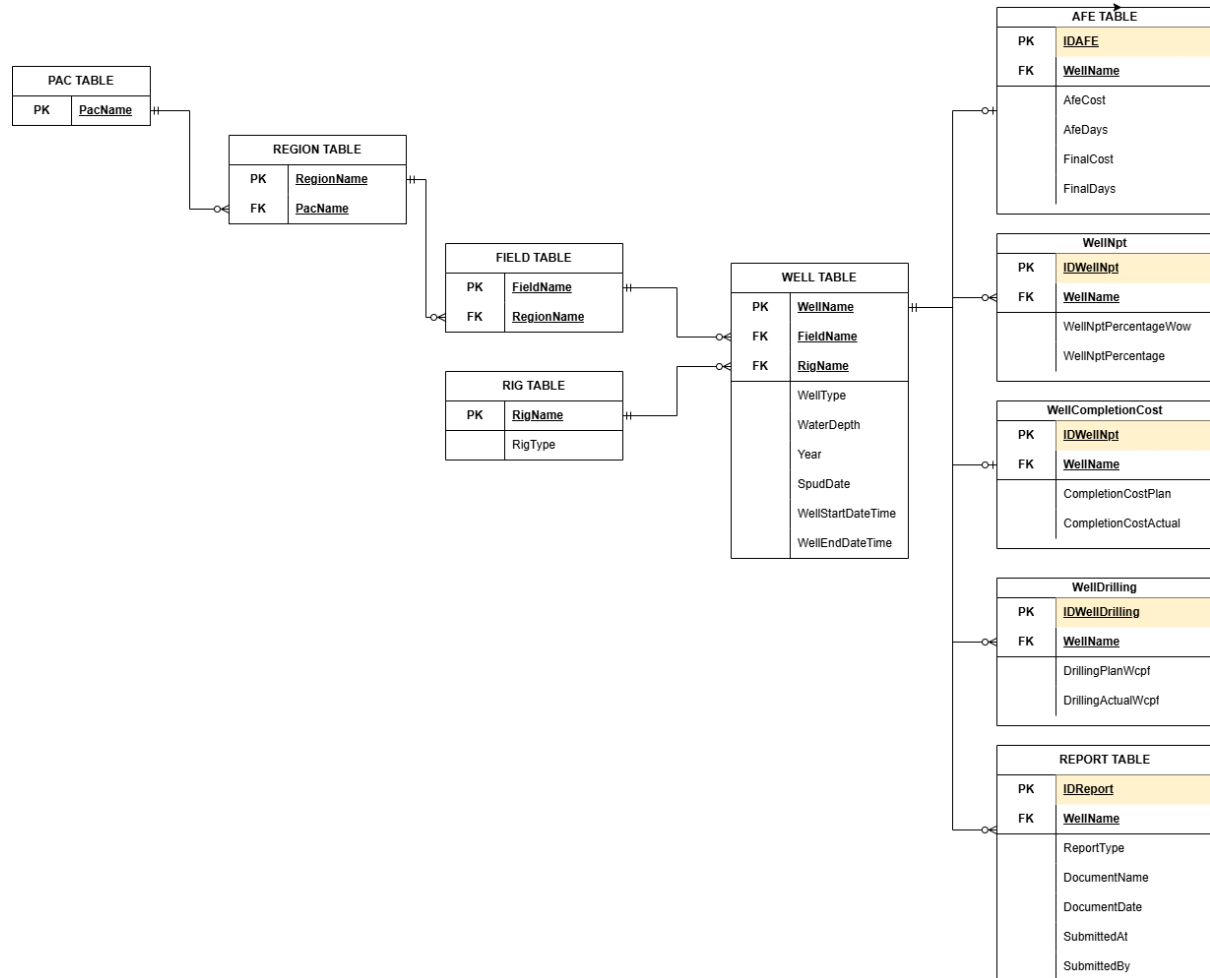
The flat table contained repeated groups of data across multiple columns and rows. Without normalization, the following problems would persist:

- **Data Redundancy:** The same PAC name, Region name, Field name, and Rig details would be repeated for every well record.
- **Update Anomalies:** Changing a region name would require updating every row that references it.
- **Insertion Anomalies:** A new PAC or Region could not be recorded without a corresponding well.
- **Deletion Anomalies:** Deleting all wells from a field would inadvertently remove the field itself.

By decomposing the flat table into related entities, each piece of information is stored exactly once, and changes propagate correctly through foreign key relationships.

3. Schema Design

The normalized schema consists of nine tables organized in a hierarchy from the broadest geographic classification down to individual well-level detail records. The tables and their relationships are described in the sections below.



3.1 PAC Table

The PAC (Production Accounting Code) Table is the top-level entity in the hierarchy. It represents the broadest organizational classification of assets.

Key	Type	Column Name	Description
PK	VARCHAR	PacName	Natural primary key. Uniquely identifies each PAC. Sourced directly from the flat table.

Design Note: PacName is a natural key derived from the source data. No surrogate key was required as the PAC name itself is unique and stable.

3.2 Region Table

The Region Table stores geographic regions. Each region belongs to exactly one PAC, establishing the first level of hierarchy below PAC.

Key	Type	Column Name	Description
PK	VARCHAR	RegionName	Natural primary key. Uniquely identifies each Region.
FK	VARCHAR	PacName	Foreign key referencing PAC.PacName. Establishes a many-to-one relationship (many regions per PAC).

Design Note: RegionName serves as the natural primary key. The relationship to PAC is one-to-many: one PAC can have many Regions.

3.3 Field Table

The Field Table stores oil and gas fields. Each field belongs to exactly one Region, forming the second level of hierarchy.

Key	Type	Column Name	Description
PK	VARCHAR	FieldName	Natural primary key. Uniquely identifies each Field.
FK	VARCHAR	RegionName	Foreign key referencing Region.RegionName. Establishes a many-to-one relationship.

Design Note: FieldName is the natural key. The relationship to Region is one-to-many: one Region can contain many Fields.

3.4 Rig Table

The Rig Table stores drilling rig information. Rigs are independent entities that are assigned to wells, so they are separated from the Well Table to avoid repeating rig attributes across multiple wells.

Key	Type	Column Name	Description
PK	VARCHAR	RigName	Natural primary key. Uniquely identifies each Rig.
	VARCHAR	RigType	Descriptive attribute indicating the type or class of the rig.

Design Note: RigName is the natural key. Separating rig data prevents RigType from being duplicated across every well that uses the same rig.

3.5 Well Table

The Well Table is the central entity of the schema. It records all individual wells and links together the Field and Rig entities. All well-level descriptive attributes are stored here.

Key	Type	Column Name	Description
PK	VARCHAR	WellName	Natural primary key. Uniquely identifies each well.
FK	VARCHAR	FieldName	Foreign key referencing Field.FieldName.
FK	VARCHAR	RigName	Foreign key referencing Rig.RigName.
	VARCHAR	WellType	Type classification of the well (e.g., producer, injector).
	DECIMAL	WaterDepth	Water depth at the well location in meters/feet.
	INTEGER	Year	The year the well was drilled or commissioned.
	DATE	SpudDate	The date drilling commenced.
	DATETIME	WellStartDateTime	Timestamp when well operations started.
	DATETIME	WellEndDateTime	Timestamp when well operations ended.

Design Note: WellName is the natural key. The Well Table acts as the central hub: all child detail tables (AFE, WellNpt, WellDrilling, etc.) reference WellName as a foreign key.

3.6 AFE Table

The AFE (Authorization for Expenditure) Table stores financial planning and actual cost data for each well. Since a well may have multiple AFEs over time, this is a child table of Well.

Key	Type	Column Name	Description
PK	INTEGER	IDAFE	Surrogate primary key. Auto-generated unique identifier because no natural key exists in the source data.
FK	VARCHAR	WellName	Foreign key referencing Well.WellName.
	DECIMAL	AfeCost	Planned/approved cost as per the AFE document.
	INTEGER	AfeDays	Planned number of drilling days per the AFE.
	DECIMAL	FinalCost	Actual final cost incurred.
	INTEGER	FinalDays	Actual number of days taken.

Design Note: IDAFE is a surrogate key introduced because the original flat table does not contain a unique identifier for each AFE record. It is highlighted in yellow/orange in the ERD to signal this design decision. The relationship to Well is many-to-one (many AFEs can relate to one well).

3.7 WellNpt Table

The WellNpt Table captures Non-Productive Time (NPT) percentage metrics per well. NPT refers to time during drilling or operations that did not contribute to progress.

Key	Type	Column Name	Description
PK	INTEGER	IDWellNpt	Surrogate primary key. Auto-generated because no natural unique identifier exists in source data.
FK	VARCHAR	WellName	Foreign key referencing Well.WellName.
	DECIMAL	WellNptPercentageWow	NPT percentage on a week-over-week basis.
	DECIMAL	WellNptPercentage	Overall NPT percentage for the well.

Design Note: IDWellNpt is a surrogate key added because multiple NPT readings can exist per well, and the source data provides no natural unique identifier for each reading. Highlighted in yellow/orange in the ERD.

3.8 WellCompletionCost Table

The WellCompletionCost Table records planned and actual completion costs for each well. Completion activities cover the work performed after drilling to prepare the well for production.

Key	Type	Column Name	Description
PK	INTEGER	IDWellNpt	Surrogate primary key. Auto-generated unique identifier.
FK	VARCHAR	WellName	Foreign key referencing Well.WellName.
	DECIMAL	CompletionCostPlan	Planned budget for completion activities.
	DECIMAL	CompletionCostActual	Actual expenditure for completion activities.

Design Note: The surrogate primary key is used here (highlighted in yellow/orange) as the source data does not provide a unique identifier for individual completion cost records. Separating this from the Well Table follows the single-responsibility principle for each entity.

3.9 WellDrilling Table

The WellDrilling Table stores drilling performance metrics, specifically the planned and actual well cost per foot (Wcpf), which is a key efficiency indicator in drilling operations.

Key	Type	Column Name	Description
PK	INTEGER	IDWellDrilling	Surrogate primary key. Auto-generated unique identifier.
FK	VARCHAR	WellName	Foreign key referencing Well.WellName.
	DECIMAL	DrillingPlanWcpf	Planned well cost per foot (Wcpf) target.
	DECIMAL	DrillingActualWcpf	Actual well cost per foot (Wcpf) achieved.

Design Note: IDWellDrilling is a surrogate key (highlighted in yellow/orange). Drilling performance data is isolated into its own table to keep the Well Table focused on core well metadata and to support scenarios where multiple drilling performance records exist per well.

3.10 Report Table

The Report Table stores document and report metadata associated with each well. This includes submission records, document dates, and report types such as daily drilling reports, end-of-well reports, or environmental submissions.

Key	Type	Column Name	Description
PK	INTEGER	IDReport	Surrogate primary key. Auto-generated unique identifier.
FK	VARCHAR	WellName	Foreign key referencing Well.WellName.
	VARCHAR	ReportType	Category or type of the report document.
	VARCHAR	DocumentName	File name or title of the associated document.
	DATE	DocumentDate	The date the document was created or is dated.
	DATETIME	SubmittedAt	Timestamp when the report was submitted.
	VARCHAR	SubmittedBy	Name or identifier of the person who submitted the report.

Design Note: IDReport is a surrogate key (highlighted in yellow/orange) because multiple reports can be associated with a single well, and the source data provides no natural unique identifier per report row. This table supports a one-to-many relationship between Well and Report.

4. Relationship Summary

The table below summarises all foreign key relationships in the schema:

Child Table	FK Column	References	Relationship
Region	PacName	PAC.PacName	Many Regions to One PAC
Field	RegionName	Region.RegionName	Many Fields to One Region
Well	FieldName	Field.FieldName	Many Wells to One Field
Well	RigName	Rig.RigName	Many Wells to One Rig
AFE	WellName	Well.WellName	Many AFEs to One Well
WellNpt	WellName	Well.WellName	Many NPT Records to One Well
WellCompletionCost	WellName	Well.WellName	Many Completion Records to One Well
WellDrilling	WellName	Well.WellName	Many Drilling Records to One Well
Report	WellName	Well.WellName	Many Reports to One Well

5. Surrogate Key Design Decisions

In several tables, no natural unique identifier could be derived from the source flat table. In these cases, a surrogate (auto-generated) integer primary key was introduced. These are visually identified by a yellow/orange highlight in the ERD. The affected tables and reasoning are as follows:

Table	Surrogate Key	Reason for Surrogate Key
AFE	IDAFE	Multiple AFE records can exist per well. No unique AFE identifier present in flat source data.
WellNpt	IDWellNpt	Multiple NPT readings per well are possible. No natural unique identifier in source data.
WellCompletionCost	IDWellNpt	Multiple completion cost entries may apply to a single well. No natural unique key available.
WellDrilling	IDWellDrilling	Multiple drilling performance records can exist per well. Source data has no unique identifier.
Report	IDReport	One well can have many associated reports. No report-level unique identifier in the flat source.

6. Data Hierarchy

The schema follows a strict top-down hierarchy that mirrors the organizational and geographic structure of oil and gas operations:

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PAC → Region → Field → Well → { AFE, WellNpt,  
WellCompletionCost, WellDrilling, Report }
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The Rig Table sits outside the main hierarchy as a reference entity, linked directly to the Well Table. This is because a rig can be assigned to wells across different fields and regions, making it an independent reference dimension rather than a hierarchical level.

7. Summary

The normalization of the flat source table into this nine-table relational schema achieves the following goals:

- Eliminates data redundancy by storing each entity (PAC, Region, Field, Rig) exactly once.
- Ensures referential integrity through foreign key constraints across all related tables.
- Introduces surrogate primary keys (highlighted in yellow/orange in the ERD) where no natural unique identifier existed in the source data.
- Separates concerns by grouping well-level detail (costs, NPT, drilling performance, reports) into dedicated child tables.
- Supports scalability, allowing new wells, AFEs, reports, or NPT records to be added without restructuring the schema.
- Follows 3NF principles ensuring each non-key attribute depends only on the primary key of its own table.

This design provides a solid foundation for downstream data analysis, reporting, and Python-based data engineering pipelines.